

GET 324 Mini-Project Report

Cloud Computing and AI Model Deployment for Engineering Applications
Group ME9 | Mechanical Engineering

Assigned Task

Binary image classification: **Tomato Leaf Mold** vs **Tomato Septoria Leaf Spot**

Project Report

Dataset Source

The dataset was sourced from the PlantVillage dataset on Kaggle. Two classes were used, Tomato Leaf Mold and Tomato Septoria Leaf Spot, totaling 2,179 training images and 544 validation images.

How to Use the Application

To use the app, open the deployed Streamlit link, upload a tomato leaf image in JPG or PNG format, and the model automatically predicts the disease class with a confidence score within seconds.

Challenges Encountered

During development, the team faced a corrupted cached weights file causing an OSError, a missing Streamlit dependency, and a file-path mismatch that stopped the deployed app from loading the trained model.

Solutions & Improvements

These were resolved by clearing the cache, installing the missing package, and running the app from the correct directory. Future improvements could include more disease classes and Grad-CAM explainability visualizations.

Model Performance Summary

Metric	Value
Base Architecture	MobileNetV2 (ImageNet pre-trained)
Training Images	2,179
Validation Images	544
Final Validation Accuracy	96.88%
Final Validation Loss	0.1144

Project Links

GitHub Repository: <https://github.com/robinson-wisdom172006/ME9>

Deployed Application: <https://groupme9.streamlit.app/>

Team Members

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